

IAC-15,A4,2,1,x29264

THE HISTORY OF THE IAA SETI PERMANENT COMMITTEE - THE EARLY YEARS

Lori A. Walton

Tigerstar Geoscience, Canada, lawalton@telus.net

Stephane Dumas

Canada, stephane_dumas@sympatico.ca

The subject of intelligent extraterrestrial life in the universe was introduced to the International Academy of Astronautics (IAA) in 1965 by Rudolph Pešek. The first IAA CETI (communication with extraterrestrial intelligence) Review Session took place in 1972 at the 23rd International Astronautical Congress (IAC) in Vienna. The IAA has held annual CETI and later SETI (search for extraterrestrial intelligence) symposia at every subsequent IAC, generating an astonishing array of presentations and papers covering technological and societal aspects of SETI. In 1974, the IAA established what is now called the SETI Committee. Members of the SETI Committee organize and conduct SETI symposia at the annual IAC meetings, select papers for publication in the IAA Journal, *Acta Astronautica* and meet annually to discuss SETI matters of importance. Since inception the SETI Committee has undertaken activities to address areas of particular concern, including the preservation of radio frequencies for SETI, designation of sites on the moon's surface for SETI activities, preparation of papers on SETI, drafting of protocols to be followed by SETI scientists in the detection, verification, announcement and response to signals from extraterrestrial civilizations and potential technical and societal aspects of a post-detection scenario. This paper will present the early history of the IAA SETI Committee with a focus on the founding of the SETI committee and the early years (1974 to 1979) of SETI Committee activities.

I. INTRODUCTION

It is truly difficult, if not impossible, to think of a committee that has a broader mandate than what is now called the SETI (Search for Extraterrestrial Intelligence) Committee* established in 1974 by its parent organization, the International Academy of Astronautics (IAA). Consider the goals and activities of the SETI Committee as stated by the International Academy of Astronautics:

1. Examine all aspects of possible future contact with extraterrestrial civilizations, with special reference to international issues and activities involving the underlying science of the study of life in the universe including:
 - relevant astrophysical and astronomical phenomena
 - the nature and distribution of biogenic elements and compounds
 - the formation of life-bearing planets
 - the origin and evolution of biological systems
 - complex life and intelligence
 - the emergence of technological civilizations

2. The SETI Committee is also charged with looking at associated space missions and technology including:
 - planetary science missions designed to search for evidence of extraterrestrial life
 - the astronomical search for extrasolar planets
 - the recovery of possible Martian microfossils on the Earth
3. The science and technology of SETI includes:
 - the development of search strategies and the underlying intellectual foundation
 - the design, development, construction and operation of telescopes and signal detection systems, studies of their future location in space or on the lunar farside
 - rejection of radio frequency interference
 - the archiving of SETI data, and analysis of search results.

And if that was not enough, the IAA also states that the SETI Committee will continue to lead in discussions of the implications of detecting extraterrestrial signals, for example, in the areas of philosophy, historical analogs, anthropology, legal, political and institutional issues, sociology, psychology, theology and interactions with the media and the education system.

Finally, the SETI Committee is expected to organize annual SETI sessions/symposia as part of the International Astronautical Congress (IAC), select peer-

* In this paper, SETI Committee is used as the short term for SETI Permanent Committee. Technically, the SETI Committee is a Permanent Committee under Commission 1 (Space Physical Sciences) under the International Academy of Astronautics.

reviewed papers of interest for publication in the scientific journal *Acta Astronautica*, work cooperatively with other groups to co-sponsor conferences, meet annually (and recently bi-annually) to discuss SETI matters and establish taskforces and workgroups to grapple with issues of importance. This includes planning for the detection of intelligent extraterrestrial life and considering the aspects and implications of a potential reply from earth.

The SETI Committee is unique in that it remains the sole long-term non-governmental international body to meet annually and to present papers, talks and examine all aspects of the search for intelligent extraterrestrial life in the universe. The value of the institutional setting for the SETI Committee within the IAA may be reevaluated from time to time; this paper may provide context for those discussions.

This paper looks at the historical setting for the founding of the IAA SETI Committee in 1974 and reviews the activities of the SETI Committee during its first five years (1974-1979). This paper is intended to be the first of a series of papers chronicling the full history of the SETI Committee, its members, its activities and achievements.

II. 1950's – CREATION OF THE INTERNATIONAL ASTRONAUTICAL FEDERATION

The roots of the SETI Committee can be traced back to the founding of the International Astronautical Federation (IAF) in 1950. Before the Second World War, rocketry enthusiasts from many countries developed contacts and exchanged information about the exciting new field. Space societies were founded, including the British Interplanetary Society, which was established in 1933. After the Second World War ended, interest in space exploration grew given advances in rocketry and propulsion. It is human nature to seek out like-minded individuals to discuss topics of mutual interest, and, inherently, during the post-war period new space groups and individuals developed local and international contacts. Interest in all aspects of space aeronautics further rallied the desire for an international venue to discuss non-military space research, space travel and related topics.

1950 - The First International Astronautical Congress

It was a somewhat improbable German, British and French triumvirate, initiated by Germany (1), who organized the first international meeting for space aeronautics, the International Astronomical Congress

(IAC), held in Paris from September 30 to October 2, 1950.

This first IAC was attended by members of rocket societies and other groups from France, Germany, Great Britain, Argentina, Austria, Denmark and Spain. At the end of the Paris meeting, it was decided that the British Interplanetary Society would organize the second IAC in London in 1951. At the same time, there was a desire for a formal federation of space societies, interplanetary societies and astronomical groups to foster astronautics and development of interplanetary flights for the non-military use of space.

The International Astronautical Federation (IAF), was thus organized in 1951 and formally founded in 1952 at the 3rd IAC in Stuttgart, Germany. The concept then, as now, is to foster dialogue between scientists around the world and support international cooperation in all space-related activities. The IAF was founded as an organization focused on the peaceful, non-military use of space. It was decided that the IAF shall hold an annual international meeting, thereby establishing the International Astronautical Congress as the IAF's annual meeting venue. A third intent was that the IAF shall promote dissemination of information between the member societies by the exchange of publications, collaboration in research, and dissemination of information to the public.

The IAF quickly became an umbrella group for groups with diverse interests in exploring space. Although technical papers and symposia in the early congresses of the IAF focused on the engineering and technological aspects of astronautics, other areas of space exploration such as space law, were examined and discussed.

The First SETI-Related Talks at the IAC

Surprisingly, it was not a radio astronomer or engineer who presented the first SETI-centric paper during the early years of the annual IAC congresses. It was an American lawyer, albeit an expert in rocket propulsion and a founding member and later President of the IAF, Andrew Haley, who presented "Space Law and Metalaw – A Synoptic View" at the 7th International Astronomical Congress in Rome in 1956. This talk, which was later published in the Harvard Law Record in 1956 (2), focused on how to project human law onto relations with extraterrestrial intelligence. Haley's talk foreshadowed the diverse array of SETI related technical and societal topics to be presented at IAC's throughout the decade and also reflected the interest in SETI by IAF members. The following year, at the 8th IAC in Barcelona, Spain in 1957, A.E. Slater presented

a talk on the probability of intelligent life evolving on a planet (3) and in 1961 R. Margaria presented a paper “On the Possible Existence of Intelligent Living Beings on Other Planets” at the 12th IAC in Washington, D.C. (4).

These SETI-related early talks at the IAC brought forward hundreds of years of historical and modern speculation about the possibility of other life in the universe.[†] Finding extraterrestrial life became closer to reality when the U.S.S.R. launched Sputnik on October 4, 1957 and humanity entered the Space Age.

III. 1960 – FOUNDING OF THE INTERNATIONAL ACADEMY OF ASTRONAUTICS (IAA)

On August 15, 1960, the 11th International Astronautical Congress opened in Stockholm, Sweden with some 800 participants. At this meeting, and after considerable debate,[‡] the International Astronautical Federation, which was a body of “member-societies”, approved the formation of an international body of distinguished individuals in the aeronautics field.

This new international body, called the International Academy of Astronautics (IAA) consisted of 45 members who appointed the open-minded and brilliant aeronautical engineer, Theodore von Kármán, as the first IAA President. Born in Hungary into a Jewish family, Karman fortuitously immigrated to the United States in 1930. Founding the IAA in 1960 was not Theodore von Karman’s only achievement; among his many contributions towards aerodynamics research was helping found the Jet Propulsion Laboratory in Pasadena, California in the 1940’s.

The IAA and another new body formed at the same time, the International Institute of Space Law (IISL), would, under the umbrella of the IAF, hold joint symposia and meetings at the annual International Astronautical Congresses. This structure remains in place today.

The IAA was never just about hard engineering and science related to astronautics and space travel. From its

[†] “Life on Other Worlds” by Steven Dick (1998) is one of several books, articles and papers on the historical search for extraterrestrial intelligent life in the universe.

[‡] An informative description of the spirited debate within member societies of the IAF regarding the founding of the IAA is provided on the International Academy of Astronautics website at iaaweb.org/content/view/43/90/

beginnings in 1960 the IAA provided a non-governmental forum through which members could contribute to advances in all aspects of the peaceful exploration of space and its benefits for humanity. Of the six commissions under the IAA’s Scientific Activities Committee, two are devoted to space policy, law and economics (Commission 5) and space and society (Commission 6). Closely aligned with its sister organization, the IAF, the IAA proposed to publish a monthly journal, *Acta Astronautica* to cover developments in space science technology in relation to peaceful scientific exploration of space and its exploitation for human welfare and progress. Papers dealing with societal aspects of space exploration would be published as well as papers on the conception, design, development and operation of space-borne and Earth-based systems.

IV. THE EARLY SIXTIES AND SETI – SETTING THE STAGE FOR THE IAA SETI COMMITTEE

When the IAA was established, the Space Age was underway and there was a growing awareness by countries that humans might detect or encounter extraterrestrial life, including intelligent life. Dick (5) describes the initiatives in the U.S.A. in the late 1950s undertaken to address the possibility of intelligent life on other planets. In July, 1958 President Eisenhower approved establishment of the Space Sciences board to advise NASA on the issue of extraterrestrial life. Two groups in the United States were dealing in the science of extraterrestrial life: the joint Armed Forces-National Research Committee and the Space Science board of the National Academy of Science. John Billingham, who would later play an influential role in the SETI Committee worked for NASA’s new Biotechnology division.

In September, 1959, the landmark paper “Searching for Interstellar Communications” was published in *Nature* (6), advocating searching for signals from extraterrestrial civilizations by monitoring the 21 cm line representing hydrogen in the radio part of the electromagnetic spectrum. Concurrently, NASA was setting up its exobiology program while Soviet radio astronomers and others were becoming interested in carrying out active searches for extraterrestrial intelligence.

The first focused search for signals from intelligent extraterrestrial civilizations took place in April-May, 1960 at Green Bank, West Virginia. As described in Shuch (7), Frank Drake, with the support of National Radio Astronomy Observatory Director Otto Struve, used the 85 foot Howard Tatel telescope at Green Bank in his off-duty time to conduct the world’s first

observational SETI experiment, Project Ozma. The modern technological search for intelligent extraterrestrial life in the universe began.

An informal conference was held a year later, in November, 1961 at the site of the Project Ozma, the National Radio Astronomy Observatory at Green Bank, West Virginia. The small group, consisting of Otto Struve (Chairman), D.W. Atchley, Jr., M. Calvin, G. Cocconi, F.D. Drake, S.S. Huang, J.C. Lilly, P.M. Morrison, B.M. Liver, C. Sagan and J.P.T. Pearman examined the broad topics to be considered in order to search for intelligent extraterrestrial life in the universe. The schematic Drake equation was adopted as a way of organizing the factors into determining the estimated number of communicative societies in the galaxy at any one time. The 1963 book *Interstellar Communication* (8) compiled the current thinking on CETI, presenting new and exciting ideas related to finding and potentially communicating with intelligent extraterrestrial life. Dyson spheres, space travel, radio transmissions, Project Ozma, masers and lasers were among the dizzying array of topics presented.

In parallel with early search efforts in the United States, radio astronomers in the U.S.S.R. were fascinated with the concept of communicating with extraterrestrial intelligent beings. In May, 1964, Soviet radio astronomers met at the Byurakan Observatory in Soviet Armenia at a conference organized by Kardashev to discuss the technical issues surrounding communication with extraterrestrial civilizations far more evolved than our own.

Also by 1964, the concept of detecting technological signatures from other civilizations had hit the front page of newspapers and popular magazines. An announcement by the U.S.S.R. that an unusual radio source labelled CTA-102 could be a beacon from another civilization was the topic of a New York Times article by journalist Walter Sullivan[§]. Although subsequently identified as a quasar, the incident raised public awareness of the scientific feasibility of detecting extraterrestrial life in the universe. The initial excitement was somewhat dashed by the first view of the seemingly lifeless surface of Mars beamed back to Earth in July, 1965 by Mariner 4. Regardless, the space race was accelerating and there was recognition that in expanding outwards into the solar system and then to the stars, that there might be life beyond the earth.

V. 1965 - A PROPOSAL FOR A CETI SYMPOSIUM

[§] New York Times, October 26, 1964

At the pivotal Stockholm meeting in 1960, a Czechoslovakian rocket scientist, Rudolf Pešek, became a member of the brand new International Academy of Astronautics and sat on its Board of Trustees. On May 15, 1965, at the 7th Session IAA Board of Trustees meeting in Paris, Rudolf Pešek proposed to his colleagues that the IAA sponsor a three day symposium on the topic of CETI (communication with extraterrestrial intelligence).** Pešek outlined a program that would cover an extensive array of physical sciences and social sciences:

1. Astronomical Perspective of Life
2. Origin of Life and Life in the Solar System
3. Evolution of Intelligence
4. Evolution of Technical Societies
5. Potential Sites for Extraterrestrial Intelligence
6. Search Problems
7. Signal Acquisition
8. CETI Impact on Mankind
9. Prospects and Recommendation for Future Research

Long before it became the norm to encourage interdisciplinary science, Pešek and other proponents of CETI within the IAA understood that the societal impact of CETI to humanity was as important as the technology and science. There was full acknowledgement that all aspects of searching for and potentially communicating with extraterrestrial life needed to be presented and discussed at the proposed symposium.

Pešek's next order of business was to find out what others thought of this new idea. Despite the dawn of the Space Age the idea of communicating with extraterrestrial intelligence was in its infancy. An international symposium where scientists could freely exchange ideas was compelling, but the IAA Board of Trustees asked Pešek to first conduct a poll of a limited number of distinguished scientists to ascertain their opinion as to whether the CETI matter was mature enough to be discussed within the IAA framework.

On July 20, 1965, Pešek posted letters to 50 scientists requesting their opinions on a symposium dealing with CETI (9). Pešek duly reported the results on September 15, 1965 at the 8th Session of the IAA Board of Trustees in Athens, Greece. Of 50 letters sent, a total of 26 replies were received in time, of which 16 (61.5%) supported the CETI concept, six (23.1%) did not think that the CETI subject matter was mature

** During this time, the topic was popularly known as CETI, as opposed to the later term SETI.

enough to be considered. Four additional replies (15.4%) were too difficult to interpret.

In light of the encouraging response, the IAA Board of Trustees approved the CETI Symposium Project and recommended that a Study Group be set up to examine the idea and to back to the Board. On September 15, 1965, IAA President Theodore von Kármán appointed the following members to the CETI Symposium Planning Study Group:

C.J. Clemedson - Sweden
V.L. Ginzburg – U.S.S.R
A.G. Haley – U.S.A
E.B. Konecci – U.S.A
R Pešek - Czechoslovakia
J.S. Shklovskii – U.S.S.R
F.L. Whipple – U.S.A

Two members were later added: S.M. Beresford and F.I. Ordway^{††}

The IAA CETI Symposium Planning Study Group met for the first and only time in Madrid, Spain on October 10, 1966 at the 17th IAC. This was the first meeting of an established international body to deal with CETI. The group came up with the following recommendations:

1. A 4 to 5 day symposium should be held to bring together individuals who are interested in CETI
2. The symposium should cover
 - Astronomical Problems
 - Biological Problems
 - Problems of communication
 - Problems of linguistics
 - Basic laws of development of civilizations
 - Impact of CETI on mankind

The IAA CETI Symposium Planning Study Group, having completed its task, submitted recommendations for consideration by the IAA Board of Trustees at the 11th Session of the Board of Trustees in Paris in April, 1967. The Board then recommended that a series of proposals be made to interested organizations and that a committee be set up to organize the proposed IAA CETI symposium, and, of course, to seek and secure financial support for such an undertaking.

The IAA Board of Trustees duly authorized the IAA President to proceed with plans to hold a symposium on the topic of CETI in 1968 and approved the President's proposal to appoint an Organizing Committee with the following members:

R. Pešek - Chairman, Czechoslovakia

^{††} Much of the information in this section is from early papers, letters, meeting minutes and articles provided by SETI Committee member Iván Almár.

C.J. Clemedson - Sweden
F.D. Drake - U.S.A
E.B. Konecci – U.S.A
B. Lowell - Great Britain
P.M. Morrison – U.S.A
F.I. Ordway – U.S.A
C. Sagan – U.S.A
J.S. Shklovskii – U.S.S.R
M. Subotowicz - Poland

In 1967 the CETI matter took on additional urgency when a unique radio source was identified by radio astronomers at the University of Cambridge in England. Although subsequently identified as a pulsar, for several weeks the group had to seriously consider whether they had discovered a signal from an extraterrestrial intelligence. In a detailed account of the incident, Alan Penny (10) describes the uncertainty over what to do, and what procedure, if any, to follow.

By 1968, the outlook was positive for the IAA's CETI symposium. On April 25, 1968, the Presidium of the Czechoslovak Academy of Sciences authorized the CETI symposium in Prague and agreed to meet the cost of lecture rooms and of interpretation. The CETI Organizing Committee met on October 18, 1968 in New York to agree on a tentative schedule of topics for the CETI symposium and set the date; Prague, September 7-12, 1969. A list of 55 potential participants was prepared.

The proposed IAA symposium on CETI never got off and running. In part, the symposium was too ambitious for the time, and, as an alternative, Nicolai Kardashev and Carl Sagan supported a smaller, more manageable Soviet-American conference, which was already far along in terms of organization. Subsequently, the first significant international conference on CETI took place September 5 to 11, 1971 at Byurakan Astrophysical Observatory in the U.S.S.R. sponsored by the Academies of Sciences of both the United States and U.S.S.R. A total of 28 Soviets, 15 Americans and 4 from other nations participated.^{††}

Shortly after the Byurakan conference, the IAA CETI Organizing Committee met in Brussels, on September 19, 1971. Rudolf Pešek reported on the Byurakan conference and proposed that the constantly delayed IAA CETI symposium scheduled for Prague be put on hiatus. Instead, The IAA CETI Organizing

^{††} The joint USA-USSR conferences continued with the second conference held in Tallin, Estonia in December, 1981 and the third and final conference of the series held in Santa Cruz, California August 5-9, 1991.

Committee agreed, in lieu of the proposed international IAA CETI symposium, to organize a half-day CETI review meeting at the 23rd IAC to be held in Vienna, 1972.

VI. 1972 – THE FIRST IAA CETI SYMPOSIUM AT THE INTERNATIONAL ASTRONOMICAL CONGRESS

By 1972, the exploration of space had matured, along with societal expectations. Apollo astronauts were walking on the moon, Pioneer 10 and its plaque were launched on March 3 to explore Jupiter and Saturn, and plans for the space shuttle were in development.

On October 14, 1972 during the 23rd International Astronautical Congress in Vienna, Austria, the tenacious Rudolf Pešek presented the inaugural talk “Survey of CETI Activities up to Present” at the first CETI symposium sponsored by the IAA, a full seven years after introducing the topic. The following five papers were presented:

Survey of CETI Activities up to Present - R. Pešek, Czechoslovak Academy of Sciences, Prague, Czechoslovakia

Astronomical Aspects of CETI - S. von Hoerner, National Radio Astronomy Observatory, Green Bank, West Virginia, U.S.A.

State of the Art in the Detection of Intelligent Extraterrestrial Signals - B.M. Oliver, Hewlett-Packard Company, Palo Alto, California, U.S.A.

Methods of Deciphering a Message from Extraterrestrial Intelligence - B.V. Sukhotin, Institute of the Russian Language, U.S.S.R. Academy of Sciences, Moscow, U.S.S.R.

State of the Art of Biological Problems - L.M. Muchin, Laboratory of Cosmic Biology, U.S.S.R. Academy of Sciences, Moscow, U.S.S.R.

VII. 1974 - A NEW IAA CETI COMMITTEE

After a second CETI Review Session in Baku, U.S.S.R. at the 24th IAC in 1973, the IAA Board of Trustees decided to set up a Standing Committee on CETI to replace the former CETI Organizing Committee. Rudolf Pešek was established as Chairman and new members were invited including John Billingham and C. Ponnampereuma from the United States. In September, 1974, the IAA President asked the

following scientists to serve as the first members of the new CETI Committee, to be followed by others.

J. Billingham, U.S.A.
C.J. Clemenson (Sweden)
V.V. Gogosov (U.S.S.R.)
A.T. Lawton (United Kingdom)
G. Marx (Hungary)
R. Pešek (Czechoslovakia)
C. Ponnampereuma (U.S.A.)
M. Subotowicz (Poland)

As before, the CETI Committee took the view that papers, talks and activities should address both the technological and sociological aspects of CETI. By the time the IAA CETI Committee was established, the first technological searches for signals from intelligent extraterrestrial civilizations had concluded and others were underway.

VII. The 1970's - THE EARLY YEARS FOR THE IAA CETI COMMITTEE

In 1974 Frank Drake and colleagues took a different approach by designing a message to be sent to extraterrestrial civilizations. On November 16, 1974 the message was transmitted to a star cluster called M33 in the constellation Hercules, by the world's largest radio telescope at Arecibo, Puerto Rico. The action had, and still has reverberations on the brand new CETI Committee. A letter was sent by astronomer Martin Ryle to the International Astronomical Union^{§§} to urge that no attempts be made to communicate with other civilizations because of possible hostile consequences (10). The question of deliberate transmissions from earth became of consequence for the CETI Committee who undertook to tackle the issue formally starting in the 1980's.

As described by Billingham and Pešek (11) 1976 was a pivotal year for science and public awareness of the scientific search for life in the universe. Organic compounds were detected in interstellar space, SETI radio searches were underway in the U.S.S.R., U.S.A. and Canada, Russian and American space probes were exploring the solar system, Project Cyclops and Project Daedalus were of interest and the term SETI as opposed to CETI was becoming more common. On July 20 and September 3, 1976, the Viking 1 and Viking 2 spacecraft landed on Mars.

^{§§} Founded in 1919, The IAU has a broad international membership of astronomers who meet every three years at the General Assembly.

At the 5th IAA CETI Review Session held in 1976 at the 27th IAC conference in Anaheim, California, U.S.A., Rudolf Pešek provided a review of all CETI activities in the IAA from 1965 to 1976 (9) At its annual meeting, also in Anaheim, the IAA CETI Committee embarked on a new initiative. Concerned about the proliferation of radio traffic, the CETI Committee adopted a resolution on radio frequency protection to allow for the search for extraterrestrial signals on specific wavelengths of the radio spectrum. The resolution was forwarded to the President of the IAA, with the request by the CETI Committee that the resolution be submitted by the IAF to the International Telecommunications Union (ITU) for consideration at the World Administrative Radio Conference (WARC) in Geneva in 1979. The resolution was approved by the IAA and IAF in 1977 and duly forwarded to the ITU. Ultimately, WARC decided to not set aside dedicated space in the spectrum for SETI activities. In 1979, SETI was discussed again at WARC and, once again, it was decided that no portion of the spectrum should be reserved for SETI but that member nations should be informed that SETI searches are taking place.

At the 28th IAC in Prague in 1977, two IAA CETI review sessions were scheduled for the first time to accommodate growing interest and the broad range of topics under discussion.

The IAA CETI Committee and the United Nations.

One of the aims of the CETI Committee, through its various names and iterations, has been to incite the interest of the United Nations in the search for intelligent life in the universe and potential communication with extraterrestrial civilizations.

The 20th session of the United Nations Committee on the Peaceful Uses of Outer Space was held in Vienna from June 20 to July 1, 1977. This session was of great importance to the IAA CETI Standing Committee since the issue of contact with extraterrestrial civilizations was discussed (12). A Czech delegation mentioned experiments in the U.S.S.R, U.S.A. and Canada to establish radio contact with extraterrestrial intelligence and asked the question as to what future steps should be taken in the field. The Austrian delegate and the Chairman of the Scientific and Technical Sub-Committee in their statements listed contacts with extraterrestrial civilizations among challenges of the future and possible future tasks for the United Nations Committee. The representative from Indonesia proposed that the Committee take up the question of SETI and that it assign the task to the Scientific and Technical Subcommittee. Of particular interest at this session was the question of deliberate messages sent from earth.

Should messages to extraterrestrial civilizations be composed under the sponsorship of an international body such as the United Nations?

Acta Astronautica – The First Special Issue

With the increase of CETI sessions at the annual IAC conferences from one to two, the number of papers on CETI produced by members of the CETI Committee and others multiplied. The first set of CETI related papers was reviewed and published in *Acta Astronautica* in 1979 (13).

Global interest in extraterrestrial life continued through the 1970's. A comprehensive Bibliography compiled by NASA in 1978 listed 1,488 items ranging from highly technical scientific papers to articles in publications such as Readers Digest, Today's Health and Theology Today (14). Members of the IAA CETI Committee authored many of the publications listed in the NASA Bibliography.

Working in cooperation with the IAU

By the late 1970's, NASA was moving forward in its Life in the Universe efforts and continued to contemplate SETI matters, detection of extrasolar planets and other items. In 1979, the IAA CETI symposiums were held at the 30th IAC in Munich, Germany, with Jill Tarter attending on behalf of John Billingham. A large Russian delegation attended this conference and the popular SETI sessions were attended by notables such as Arthur C. Clarke.

Also in 1979, the International Astronomical Union sponsored a session on "Strategies for the Search for Life in the Universe" during the August, 1979, 17th General Assembly of the International Astronomical Union in Montreal. Encouraged by the large number of astronomers interested in the topic, the 18th General Assembly of the IAU in August 1982 at Patras, Greece endorsed the proposal of the Executive Committee of the IAU to establish a commission devoted to the search for extraterrestrial life. In the spirit of encouraging international cooperation, Rudolf Pešek and John Billingham, as Co-Chairs of the IAA CETI Committee assisted the new IAU Commission 51 and Michael Papagiannis was appointed an official representative of the IAU to the IAF and subsequently accepted membership in the IAA CETI Committee. In 1984 the first Bioastronomy Commission Symposium took place in Boston in 1984.^{***}

^{***} The IAU Bioastronomy Conferences sponsored by Commission 51 of the IAU were triennial events. The Bioastronomy Conference was in Puerto Rico in

In 1980, Rudolf Pešek, stepped down as Chairman of the CETI Standing Committee and John Billingham from the United States assumed the Chair role, thus ending the first era of the IAA CETI Committee.

VIII. CETI COMMITTEE POST 1980 – THE WORK CONTINUES

The second paper in this series on the History of the SETI Committee will examine the SETI Committee activities from 1980 to 2000. This period covers the drafting of the Post Detection Protocol and a position paper to address the issue of transmissions from earth, preparation for key presentations on SETI and Protocols, and collaborations with other SETI committees and conferences. Significantly, by the end of the 1990s, the first exoplanets were identified, and, suddenly, the concept of extraterrestrial life became closer to reality. The third paper in the series will examine the IAA SETI Committee activities in the 22nd century, from year 2000 to present, and how the SETI committee views its future plans.

IX. CONCLUSION

When the IAA SETI Committee was formally established in 1974, the concept of finding life in the universe, let alone “intelligent” life was academic. Through the decades, humanity has viewed a diversity of worlds via the images and data sent back by probes examining our solar system. A staggering number of exoplanets have been identified and through biological studies on the origin of life and examination of all life on earth, it is now recognized that life can exist in extreme environments. Searching for biosignatures on other worlds is moving forward and with it is a general awareness that technosignatures indicating intelligent life may be detected.

In this current context, the SETI Committee, now in its 42nd year, is the only international body that meets annually to hold meetings and organize symposia and the exchange of information on all aspects of the search for intelligent extraterrestrial life in the universe. It has functioned with the strong support of the IAA and its members.

ACKNOWLEDGEMENTS

We would like to thank SETI Committee Chair Claudio Macconne for suggesting this topic. We also

2007. Subsequently, Commission 51 combined the Bioastronomy conference series with the Origins of Life Conferences and SETI related talks were phased out.

gratefully acknowledge fellow SETI Committee member Alan Tough, who passed away in 2012 for arranging the Billingham Cutting Edge Lecture series in conjunction with the IAA SETI sessions at the International Astronautical Congresses. SETI Committee member Paul Shuch maintains a website at avsport.org/IAA, which together with the IAA website provides a repository for historical papers and documents. The authors would especially like to thank Ivan Almar for his contribution of historic papers pertaining to the IAA SETI Committee. L. Walton would like to thank Jill Tarter and Chris Neller at the SETI Institute for their support for the archiving of SETI-related documents held at the SETI Institute.

REFERENCES

1. Skoog, A. Ingemar, 2010, The Swedish Interplanetary Society (1950 – 1969) and the Formation of IAF and IAA. Manuscript, IAC-10-E4.1.04, International Astronomical Congress, Prague.
2. Haley, A.G., 1956, Space Law and Metalaw – A Synoptic View. Harvard Law Record 23, Nov 8, 1956
3. Slater, A.E. 1958, The Probability of Intelligent Life Evolving on a Planet. Proceedings of the 8th International Astronautical Congress, Barcelona, Spain, 1957, Springer-Verlag, Vienna, p. 395-402.
4. Margaria, R., 1961, On the Possible Existence of Intelligent Living Beings on Other Planets. Proceedings of the 12th International Astronautical Congress, Washington, D.C., Makemson, M.W. and Baker, R.M.L. (Eds.), Springer-Verlag, Vienna, 556-563.
5. Dick, Steven J. and Strick, James E., 2004, The Living Universe: NASA and the Development of Astrobiology. Rutgers University Press, New Jersey, 308 p.
6. Cocconi, G. and Morrison, P., 1959, Searching for Interstellar Communications. Nature, vol. 184, No. 4690, p. 844-846.
7. Shuch, Paul H., 2011, Project Ozma: The Birth of Observational SETI. In Searching for Extraterrestrial Intelligence, The Frontiers Collection, Shuch, Paul H. (Ed.), Springer-Verlag Berlin Heidelberg, p. 13-18.

8. Cameron, A.G.W. (Ed.), 1963, *Interstellar Communication – A Collection of Reprints and Original Contributions*. W.A. Benjamin, Inc., New York, Amsterdam, 320 p.
9. Pešek, R, 1979, *Activities of the IAA CETI Committee from 1965 – 1976 and CETI Outlook*. *Acta Astronautica*, vol. 6, p. 3-9
10. Penny, Alan, 2013, *The SETI Episode in the 1967 Discovery of Pulsars*. *European Physical Journal H*, vol. 38, issue 4, pp. 535-547.
11. Billingham, J. and Pešek, R. (Eds.), 1979, *Communication with Extraterrestrial Intelligence*. Pergamon Press, Oxford
12. United Nations (1977), *Messages to Extraterrestrial Civilizations* (Note by the Secretariat). Committee on the Peaceful Uses of Outer Space, Document A/AC.105/206, October 18, 1977, United Nations, New York.
13. *Acta Astronautica*, Special Issue (1979) CETI: Communication with Extraterrestrial Intelligence, vol. 6, 102.
14. Mallove, Eugene F., Connors, Mary M., Forward, Robert L. and Paprotny, Zbigniew, 1978, *A Bibliography on the Search for Extraterrestrial Intelligence*. NASA Reference Publication 1021, Scientific and Technical Information Office, 132 p.